

**Instructions:**

Please read the following instructions carefully before submitting the assignment:

You need to use MS Word document to prepare and submit the assignment on VU-LMS.

It should be clear that your assignment will not get any credit if:

- The assignment is submitted after the due date.
- The assignment is not in the required format (doc or docx).
- The submitted assignment does not open or file is corrupt.
- Assignment is copied (partial or full) from any source (websites, forums, students, etc.).

**Objective:**

To enhance the learning capabilities of the students about:

- Network Topology
- Protocol Models
- Noisy Channels and Data Rate Limits

### **Question No. 1**

Life-Care Hospital is upgrading its digital infrastructure. The Emergency ward contains 15 monitoring devices. To ensure maximum reliability, every device is connected to every other device via a dedicated physical link. The Pharmacy contains 25 computers connected to a central layer-2 switch. The connection between the Ward and the Pharmacy has a bandwidth of 2 MHz and a signal-to-noise ratio of 15.

Considering the above scenario, you are required to answer the following questions:

|    |                                                                                                                                                                              |   |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 1. | Identify the Topology used in the Emergency ward and calculate the total number of cables required to connect all 15 devices.                                                | . |
| 2. | In the Pharmacy, if one computer sends a message to another, identify the central device and state if the data is sent to all 25 computers or only the destination computer. | . |
| 3. | Calculate the Shannon Capacity (maximum bit rate) of the link between the Ward and the Pharmacy in Mbps.                                                                     | . |

### **Question No. 2**

The hospital uses the TCP/IP protocol suite. Analyze the table below; the information in Column Y is currently incorrect. Write the correct letter from Column Y into Column Z for each layer.

| <b>X</b>          | <b>Y</b>                                                          | <b>Z</b> |
|-------------------|-------------------------------------------------------------------|----------|
| Application Layer | <b>A.</b> Bit-level transmission over the medium                  |          |
| Network Layer     | <b>B.</b> Provides services and access to the end-user            |          |
| Transport Layer   | <b>C.</b> Responsible for source-to-destination (packet) delivery |          |
| Physical Layer    | <b>D.</b> Responsible for Process-to-Process (Segment) delivery   |          |

**Best Of Luck!**